FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Advanced)-2019

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. Use $\int_0^4 F dt = mv - m(5)$

$\text{work}_F = \frac{1}{2}mv^2 - \frac{1}{2}m(5)^2$

2. D

Sol. $\tan\theta = \frac{a}{g} = \frac{y_1 - y_2}{4}$

$\therefore y_1 - y_2 = 0.5$ .....(i)

Volume of water is

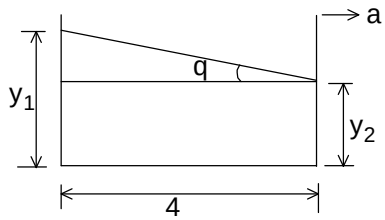
$\frac{4(y_1 + y_2)}{2} \times 3 = 18$

$\therefore y_1 + y_2 = 3$ .....(ii)

$\therefore y_2 = \frac{2.5}{2}$

$\therefore$ speed of liquid coming from orifice is

$= at + \sqrt{2gy_2} = 10\,m/s$



3. C

Sol. $T_P = 40$ N

$T_Q = 30$N

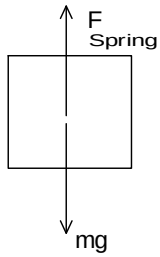
$T_P - T_Q = 10$ N

4. BD

Sol. When a displacement wave strikes a fixed end, it gets inverted after reflection and when a pressure wave strikes a fixed end, compression pulse gets reflected as compression pulse.

5. ABD

Sol. Speed will be maximum when the block passes through the equilibrium position.



$F_{spring} = mg$

$\therefore$ A is correct

(B) From conservation of energy

$$mg(\ell + x) = \frac{1}{2}Kx^2 + \frac{1}{2}mv_{max}^2 \qquad (1)$$

Where $x = \frac{mg}{K} = \frac{mg}{4mg}\ell = \frac{\ell}{4}$

putting $x = \frac{\ell}{4}$ in equation (1)

$$V_{max} = \frac{3}{2}\sqrt{gl}$$

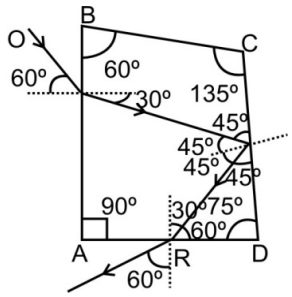
(C) is incorrect as in equilibrium compression is $\ell/4$ which is more than $\frac{\ell}{8}$

(D) Time taken is $\frac{T}{4} = \frac{1}{4}2\pi\sqrt{\frac{m}{K}}$

$\therefore$ D is correct

6. ABC

Sol. Applying Snell's law at face AB, we get



$1 \times \sin 60^\circ = \sqrt{3} \times \sin r$

$\frac{\sqrt{3}}{2} = \sqrt{3}\sin r$ or $\sin r = \frac{1}{2}$

$r = \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$

$\sin\theta_c = \frac{1}{\sqrt{3}}$

(where, θ_c is the critical angle)

$\theta_c = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ or $\theta_c \approx 35^\circ$

From geometry, angle of incidence at Q is 45º. At face CD angle of incidence is greater than θ_c. Therefore, the rays gets totally internally reflected at face CD. Hence, option (a) is correct. From geometry, angle of incidence at R is 30º. At face AD angle of incidence is less than θ_c

So, the ray comes through face AD. Hence option (b) is correct.

Applying Snell's law at face AD, we get

$e = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = 60^\circ$

The angle between the incident ray and the emergent ray is 90º.

Hence, option (c) is correct and option (d) is incorrect.

7. AD

Sol. $W_{A\to B} = nRT_1 \ln 2$

$$W_{A\to B} = \frac{U_1 - U_0}{2} \ln 2$$

$$\frac{dv}{dT} = 2R$$

$$dv = 2RdT$$

$$C_V = 2R$$

$$C_P = 3R$$

8. BCD

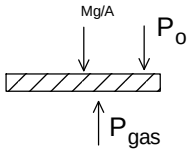
Sol. Current through resistances remains constant.

9. B

Sol. $Q = nC_P\Delta T = C_P\left(\frac{P\Delta V}{R}\right)$

10. A

Sol.



SECTION – C

11. 8

Sol. $\left(P_o + \frac{4S}{R}\right)R^3 = \left(0.5P_o + 0.8\times\frac{4S}{R}\right)\left(\frac{5R}{4}\right)^3$

$$P = \frac{96S}{R}$$

12. 4

Sol. $L_{min} = \frac{C}{4f} = 16 \text{ cm}$

so resonance for length 16 cm, 48 cm, 80 cm i.e 3 cases.

13. 5

Sol. $V_S^2 = V_R^2 + V_C^2 + 2V_RV_C\cos\phi$

$$i = \frac{V_R}{R}$$

$$P_C = i\,V_C\cos\phi$$

14. 3

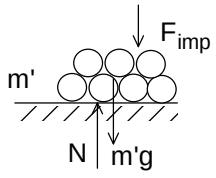
Sol. each falling link collides with the fallen links with a relative velocity

$\vec{v}_{imp} = -v\hat{j}$

$\vec{f}_{imp} = \vec{v}_{rel}\frac{dm^1}{dt}$

$\vec{f}_{imp} = -\mu v^2\hat{j}$

from equation on heap



$N - m'g + f_{imp} = 0$

$N = \mu\left(v^2 + gy\right)$

$= 2\left(1^2 + 10 \times 0.05\right)$

$= 2\left(1 + 0.5\right)$

$= 3N$

15. 5

Sol. $P'_M = 2P_2 + P_M$

$P_2 = \dfrac{1}{f_2} = \dfrac{1}{4R}$

$-\dfrac{1}{f'_m} = 2 \times \dfrac{1}{4R} + \dfrac{2}{R} = \dfrac{5}{2R}$

$f'_m = -\dfrac{2}{5}R$

$|h| = 2|f'_m|$

$h = 2 \times \dfrac{2R}{5} = 4$

$R = 5m$

16. 5

Sol. $E = \dfrac{1}{2}mv^2 + \dfrac{1}{2}kx^2 + \dfrac{q^2}{2c}$

$e = Bv\ell = \dfrac{q}{c}$

$E = \dfrac{1}{2}\left(m + B^2\ell^2c\right)v^2 + \dfrac{1}{2}kx^2$

as E = constant

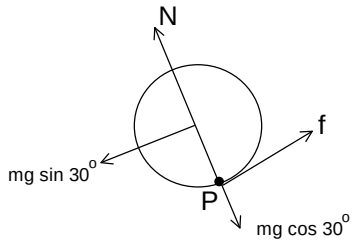
$\dfrac{dE}{dt} = 0$

$\dfrac{dv}{dt} = -\left(\dfrac{k}{m + B^2\ell^2c}\right)x$

ω = 20 rad/sec

17. 5

Sol. Angular momentum about P = $\tau\Delta t = \left(mg\sin 30^o\right)R(1)$



18. 3

Sol. $\because$ P.E., $U = -\vec{\mu}.\vec{B} = (-)\dfrac{\left(I_2\pi r^2\right)\mu_o I_1 R^2}{2\left(R^2 + x^2\right)^{\frac{3}{2}}}$

$$F = (-)\frac{du}{dx} = \frac{3\mu_o \pi R^2 r^2 I_1 I_2 x}{2\left(R^2 + x^2\right)^{\frac{5}{2}}}, \therefore N = 3$$

19. 3

Sol. $\because \mu = \dfrac{\sin\left(\dfrac{A+\delta_m}{2}\right)}{\sin\left(\dfrac{A}{2}\right)} \Rightarrow \dfrac{2}{\sqrt{3}} = \dfrac{\sin\left(\dfrac{A+\frac{A}{2}}{2}\right)}{\sin\left(\dfrac{A}{2}\right)}$

$$\Rightarrow \frac{2}{\sqrt{3}} = \frac{\sin\left(\frac{3A}{4}\right)}{\sin\left(\frac{A}{2}\right)}, \text{Let} \quad \frac{A}{4} = \theta$$

$$\Rightarrow \frac{2}{\sqrt{3}} = \frac{\sin 3\theta}{\sin 2\theta} = \frac{3\sin\theta - 4\sin^3\theta}{2\sin\theta.\cos\theta} = \frac{3 - 4\sin^2\theta}{2\cos\theta}$$

$$\Rightarrow 4\sqrt{3}\cos^2\theta - 4\cos\theta - \sqrt{3} = 0$$

$$\Rightarrow \cos\theta = \frac{1 \pm 2}{2\sqrt{3}} = \frac{\sqrt{3}}{2}, \left(-\frac{1}{2\sqrt{3}}\right)$$

$$\Rightarrow \theta = 30^o \Rightarrow \frac{A}{4} = 30^o = \frac{\pi}{6}$$

$$\Rightarrow A = \frac{2\pi}{3}, \therefore C = 3$$

20. 1

Sol. For maxima at P, path Diff. $= n\lambda$, $n =$ integer

$$\Rightarrow \left(s_2P - s_1P\right) - (\mu - 1)t = n\lambda \Rightarrow y\frac{d}{D} - (\mu - 1)t = n\lambda$$

$$\Rightarrow n\lambda = 10^{-6}m \Rightarrow n = \frac{10^{-6}}{\lambda}$$

$$n^{max} = \frac{10^{-6}}{4000 \times 10^{-10}} = 2.5 \,\&\, n^{min} = \frac{10^{-6}}{7000 \times 10^{-10}} = 1.42$$

$\therefore n = 2, \therefore$ only one wavelength will give maxima at P for $n = 2$

SECTION – D

21. 00001.53

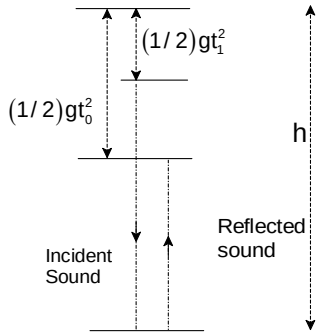
Sol. The centre of mass of the rod is at distance $\dfrac{(y-x)}{2}$ from the brim.

$$\left(\rho_{al} gV - \rho_w g\frac{V}{2}\right)x = mg\frac{(y-x)}{2}$$

$$\Rightarrow \frac{4}{x} = 1.53$$

22. 01057.50

Sol. Let the sound observed by the parachutist at t_0 = 12s be produced at t_1s. Velocity of source at the instant of emitting sound = gt_1 and velocity of observer at the instant of observing same sound = gt_0. Hence the relation between apparent frequency f' and original frequency f will be



$$f' = f\left(\frac{v + gt_0}{v - gt_1}\right)$$

Here f = 800 Hz, g = 10 m/s^2, v = 330 m/s
T_0 = 12s and f' = 800 + 700 = 1500 Hz
Putting these, we get
T_1 = 9s
Now the distance traveled by sound in $(t_0 - t_1)$ sec

is $v(t_0 - t_1) = \left(h - \frac{1}{2}gt_0^2\right) + \left(h - \frac{1}{2}gt_1^2\right)$

Putting the values, we get
H = 1057.5 m

23. 00772.00

Sol. Rate of formation of nuclei at time t is

$\frac{dN}{dt} = \alpha - \lambda N$, where N = number of nuclei at that time

$$N = \frac{\alpha}{\lambda}\left(1 - e^{-\lambda t}\right)$$

Number of nuclei disintegrated in time t, $N_d = \alpha t - \frac{\alpha}{\lambda}\left(1 - e^{\lambda t}\right)$

$\therefore$ Energy released till time t, $E = E_0 N_d = E_0\left[\alpha t - \frac{\alpha}{\lambda}\left(1 - e^{-\lambda t}\right)\right]$

20% of this energy is used for heating water,

$$0.2E_0\left[\alpha t - \frac{\alpha}{\lambda}\left(1 - e^{-\lambda t}\right)\right] = ms\Delta\theta$$

$$\Delta\theta = \frac{0.2E_0\left[\alpha t - \frac{\alpha}{\lambda}\left(1 - e^{-\lambda t}\right)\right]}{ms}$$

At $t = T_{1/2}$

$$\Delta\theta = \frac{0.2E_0\left[2\alpha T_{\frac{1}{2}} - \frac{2\alpha}{\lambda}\left(1 - \frac{1}{2}\right)\right]}{ms}$$

$$\Delta\theta = \frac{0.2E_0[0.386]}{ms} 772K$$

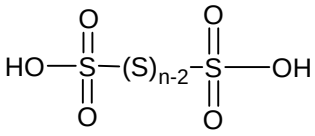
Chemistry

PART – II

SECTION – A

24. B

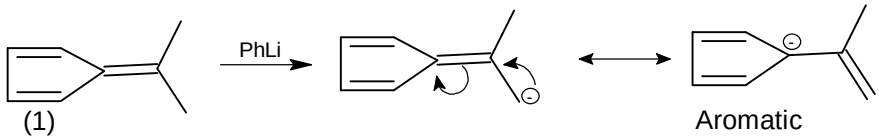
Sol.



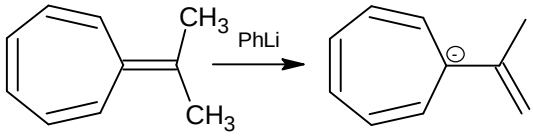
No. of S-S bond = (n-1)

25. C

Sol.



(1) Aromatic



(2) Anti Aromatic

26. C

Sol. $pv^{5/2} = const$; $n = 5/2$ (Polytropic process)

no. of moles $= 1$

Cv of Ar $= \frac{3}{2}R$

$q = nC \cdot \Delta T$

$= n\left(Cv + \frac{R}{1-n}\right)\Delta T$

$= 1\left(\frac{3}{2}R + \frac{R}{1-5/2}\right) \times 26$

$= \frac{5}{6}R \times 26$

$= \frac{5}{6} \times 8.314 \times 26 = 180J$

27. ABC

Sol. (a) Mechanical equilibrium
(b) Thermal equilibrium
(c) Chemical equilibrium

28. ABC

Sol. $A_2O_n + MnO_4^- \xrightarrow{H^+} AO_3^- + Mn^{+2}$

$Eq.\ of\ A_2O_n = Eq.\ of\ MnO_4^-$

$1.5 \times 10^{-3} \times (10 - 2n) = 60 \times 0.03 \times 10^{-3} \times 5$

$no.\ of\ moles \times (n-factor)$

$\therefore n = 2$

$formula = A_2O_2$

$Empirical\ formula = AO$

29. ABCD

Sol. ABCD

I=Trans form but both methyl on equational position

IV=III=cis form one methyl on axial position and one on equational position

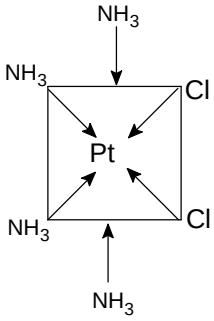
II=Trans form but both methyl on axial position

30. AB

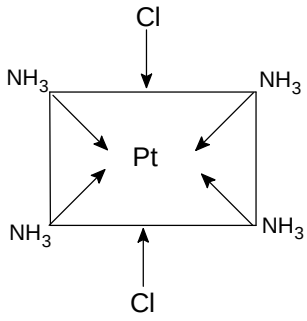
Sol. In hyper conjugation σ electron interact with vacant orbital.

31. ACD

Sol. (A)



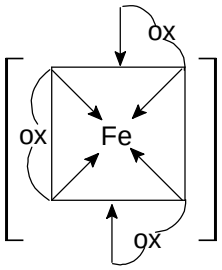
cis



trans

Plane of symmetry optically in active Plane of symmetry optically in active

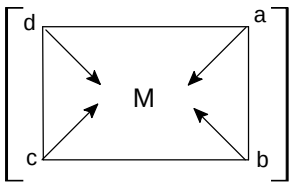
(B)



-3

- Optically active
- no geometrical isomerism

(C)



$\pm n$

Square planer
Plane of symmetry
Optically inactive
- GI occurs

(D) If ligand is optically active then complex is optically active.

32. D

Sol. In polymerisation isomerism; ratio of metal and different ligand should be same.

33. B

SECTION – C

34. 8

Sol. $\left[PtCl_2(NH_3)_4\right]\left[PtCl_2(SCN)_4\right]$ c

$\left[PtCl(SCN)(NH_3)\right]\left[PtCl(SCN)_3\right]$

↓

ambidentate ligand

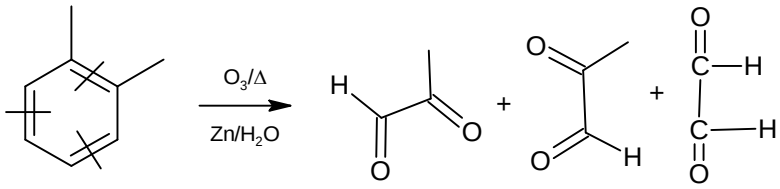
so two isomer possible

Total $\left(SCN^-\right)$*ligand* $= 4$

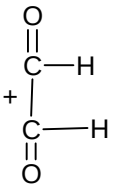
Total isomer $= 4 \times 2 = 8$

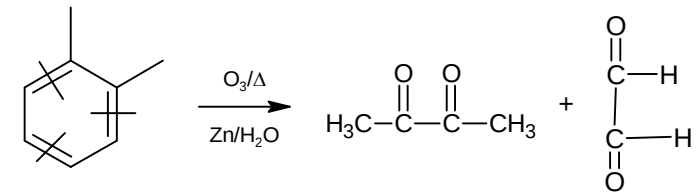
35. 3

Sol.

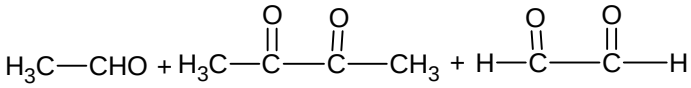


$= 2CH_3CHO +$



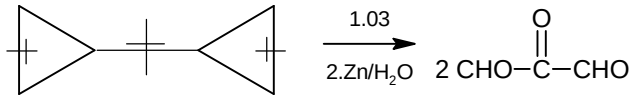


So, no. of different pdt are



36. 3

Sol.



(X)

No. of π bond = 3

37. 6

Sol.

$10 \rightarrow 4$
$9 \rightarrow 4$
$8 \rightarrow 4$
$7 \rightarrow 4$
$6 \rightarrow 4$
$5 \rightarrow 4$

} *Total* 6 *spectral lines*

38. 2

Sol. $\Delta T_f = K_f m = k_f \times \frac{15/m}{100} \times 1000$

$1 = k_f \times \frac{15}{m} \times 10 \quad ---(1)$

if mass of solute $= 30g$

$\Delta T_f = K_f \times \frac{30/m}{100} \times 1000$

$= 2 \times k_f \times \frac{15}{m} \times 100$

$= 2 \times 1 = 2$

39. 4

Sol. $HA + NaOH \rightarrow NaA + H_2O;\ \Delta H = -6900\, cal/eg$

$HB + NaOH \rightarrow NaB + H_2O;\ \Delta H = -2900\, cal/eg$

Let x eq of NaOH react with HA and y eg NaOH react with HB

$x + y = 1$

$-6900\, x(x) - 2900 \times y = -3900$

$\therefore x = 1/4$

$y = 3/4$

$x : y = 1 : 3$

$x + y = 1 + 3 = 4$

40. 1

Sol. SO_4^{-2} ion is present in solution; no electrolysis of SO_4^{-2}

$[SO_4^{-2}]$=0.1 M

Decimolar concentration of $SO_4^{-2} = 1$

41. 6

Sol. $\frac{r_{Mg^{+2}}}{r_{S^{-2}}} = \frac{0.65}{1.84} = 0.35$

Co-ordination no.=4

$\frac{r_{Mg^{+2}}}{r_{O^{-2}}} = \frac{0.65}{1.4} = 0.464$

Co-ordination no.=6

$\frac{r_{Cs^+}}{r_{Cs^-}} = \frac{1.69}{1.81} = 0.933$

Co-ordination no.=8

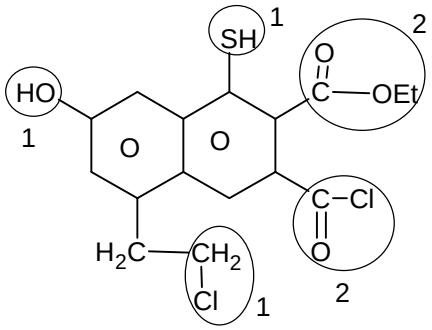
	Co − ordination no.
$0.225 < \frac{r+}{r-} < 0.414$	4
$0.414 < \frac{r+}{r-} < 0.732$	6
$0.732 < \frac{r+}{r-} < 1$	8

42. 7

Sol. *a*, *c*, *d*, *g*, *j*, *l*, *m*

43. 7

Sol.



SECTION – D

44. 00002.68

Sol. $Density = \frac{Z \times M}{N_A \times a^3}$

$a^3 = \frac{Z \times M}{N_A \times d}$

$Distance\ between\ K^+\ and\ F^- = \frac{a}{2}$

45. 00040.00

Sol. Rate after 10 minutes = $\frac{-(6-8)}{10} = \frac{2}{8}$

Rate after 20 minutes = $\frac{-(4-6)}{10} = \frac{2}{8}$

Reaction of 1st order

$A_t = A_o - kt$

$0 = 8 - k \times t$

$t = \frac{8}{k}$

$6 = 8 - k \times 10$

$\therefore\ k = \frac{2}{10}$

$t = \frac{8}{k} = \frac{8}{\frac{2}{10}} = 40\ min$

46. 00000.11

Sol. no of equivalent of Al = $\frac{0.5 \times 40 \times 60}{96500}$

No of moles of Al = $\frac{0.5 \times 40 \times 60}{96500} \times \frac{1}{3}$

Mass of Al deposited = $\frac{0.5 \times 40 \times 60}{96500} \times \frac{27}{3} = 0.11\,g$

Mathematics

PART – III

SECTION – A

47. A

Sol. $A = \frac{1}{16}\left(1+\frac{1}{4}+\frac{1}{4^2}+.....\right) = \frac{\frac{1}{16}}{1-\frac{1}{4}} = \frac{1}{12}$ sq. unit

48. C

Sol. Area of required region is half of the area of circle.

49. C

Sol. $x = 1+h, \quad h \to 0$

$$\lim_{h\to 0} \frac{\left(\log(2+h)-\log 2\right)\left(3.4^h-3-3h\right)}{\left((8+h)^{\frac{1}{3}}-(4+3h)^{\frac{1}{2}}\right)(-\sin\pi h)}$$

$$\lim_{h\to 0} \frac{3\left(\frac{\frac{1}{2}\log\left(1+\frac{h}{2}\right)}{\frac{4}{2}}\right)\left(\frac{4^h-1}{h}-1\right)}{\left\{\frac{(8+h)^{\frac{1}{3}}-8^{\frac{1}{3}}}{h}-\frac{3(4+3h)^{\frac{1}{2}}-4^{\frac{1}{2}}}{3h}\right\}\left\{-\pi\frac{\sin\pi h}{\pi h}\right\}}$$

$$= \frac{3\times\frac{1}{2}\times(\log 4-1)}{\left(\frac{1}{3}.\frac{1}{4}-3\times\frac{1}{2}\times\frac{1}{2}\right)(-\pi)} = \frac{\frac{3}{2}(\log 4-1)}{4\frac{\pi}{3}}$$

$$= \frac{9}{4\pi}(\log 4-1)$$

50. C

Sol. $\vec{r}-\vec{c} \parallel \vec{b} \Rightarrow \vec{r} = \vec{c}+\lambda\vec{b}$

$\vec{r}.\vec{a} = 0 \Rightarrow \lambda = -\frac{\vec{a}.\vec{c}}{\vec{a}.\vec{b}} = \frac{-(8+7)}{2+0+1} = -5$

$\vec{r} = \vec{c}+\lambda\vec{b} = \left(4\hat{i}-3\hat{j}+7\hat{k}\right)-5\left(\hat{i}+\hat{j}+\hat{k}\right)$

$\vec{r} = -\hat{i}-8\hat{j}+2\hat{k}$

$\therefore x+y+z = -7$

51. C

Sol. $\tan\omega = \pm\left|\frac{m_1-m_2}{1+m_1m_2}\right| = \pm\left|\frac{a}{b}\frac{\left(\tan\theta+\frac{1}{\tan\theta}\right)}{1-\frac{a^2}{b^2}}\right|$

$$\cot\theta = \pm\frac{\left(a^2-b^2\right)\sin\theta.\cos\theta}{a.b}$$

$$\therefore \frac{2\cot\theta}{\sin 2\theta} = \pm\frac{\left(a^2-b^2\right)}{a^2}\left(\frac{a}{b}\right) = \pm\frac{e^2}{\sqrt{1-e^2}}$$

52. A

Sol. The line $x - y + a = 0$ tangent at P. then $|PR - PQ|$ will be maximum $(= RQ)$

53. ACD

Sol. focus $(6,-3)$ and equation of tangent at vertex is $5x + 3y = 15$.

$$\therefore \text{Length of latus rectum} = 4\frac{|30-9-15|}{\sqrt{34}} = \frac{24}{\sqrt{34}}$$

54. BD

Sol. As $x \to \frac{\pi}{2}^-$, $(\cot x$ is positive $(\text{being in I quadrant}))$

thus $[\cot x] = 0$ $\lim\limits_{x\to\frac{\pi}{2}^-} f(x) = e^0 = 1$

$f(x) = e^{[\cot x]}$ as $\cot x$ is negative in 2nd quadrant

and $\cot\frac{\pi}{2} = 0 \Rightarrow [\cot x] = -1$

$\lim\limits_{x\to\frac{\pi}{2}^+} f(x) = \frac{1}{e}$

55. C

Sol. $A \geq G \Rightarrow \frac{\cot\frac{A}{2}+\cot\frac{C}{2}}{2} = \cot\frac{B}{2} \geq \sqrt{3}$

$\cot\frac{B}{2} \geq \sqrt{3}$

56. C

Sol. $\tan\frac{B}{2} \leq \frac{1}{\sqrt{3}} \Rightarrow B \leq \frac{\pi}{3}$

SECTION – C

57. 1

Sol. Ellipse in internally tangent to the circle at point of intersection.

58. 6

Sol. Let $\left(x+\frac{1}{x}\right)^3 = a$ and $x^3 + \frac{1}{x^3} = b$

$$f(x) = \left|\frac{a^2-b^2}{a+b}\right| = |a-b| = \left|\left(x+\frac{1}{x}\right)^3 - x^3 - \frac{1}{x^3}\right|$$

$$= 3\left|x+\frac{1}{x}\right| \geq 6.$$

59. 0

Sol. use property

60. 4

Sol. $H = xy = c^2, S \equiv (x-\lambda)^2 + (y-\beta)^2 = r^2$ let $P\left(ct, \frac{c}{t}\right)$

$$S_1 = t_1^2 + t_2^2 + t_3^2 + t_4^2 = 2\frac{(\alpha^2 - \beta^2 + \gamma^2)}{c^2}$$

$$S_2 = \frac{1}{t_1^2} + \frac{1}{t_2^2} + \frac{1}{t_3^2} + \frac{1}{t_4^2} = 2\frac{(\beta^2 - \alpha^2 + \gamma^2)}{c^2}$$

$$\lambda = \frac{c^2}{r^2}(S_1 + S_2) = \frac{c^2}{r^2} \times \frac{4r^2}{c^2}$$

61. 5

Sol. Let y = cotx; $3y^2 + 8y + 3 = 0$, y_1, y_2 be roots

$y_1 + y_2 = -\frac{8}{3}, y_1y_2 = 1$

$y = \cot x$ is a bijection on $(0, \pi) \to R$

and also $(\pi, 2\pi) \to R$

$\cot x < 0$ when $x \in \left(\frac{\pi}{2}, \pi\right) \cup \left(\frac{3\pi}{2}, 2\pi\right)$

$\therefore x_1, x_2 \in \left(\frac{\pi}{2}, \pi\right)$ & $(x_3, x_4) \in \left(\frac{3\pi}{2}, 2\pi\right)$

$y_1 = \cot x_1 = \cot x_3$, $y_2 = \cot x_2 = \cot x_4$

$y_1y_2 = 1 \Rightarrow \cot x_1 \cot x_2 = \cot x_3 . \cot x_4 = 1$

$\therefore x_1 + x_2 + x_3 + x_4 = 5\pi$

62. 3

Sol. $B = \text{adj}(2A) \Rightarrow |B| = |2A|^{n-1}$

$1024 = 4^{n-1} \times 2^{n(n-1)}$

$2^{10} = 2^{n^2+n-2}$

$n = 3$

63. 6

Sol. Let (h, k) lies on hyperbola

$$\frac{h - x_1}{2} = \frac{k - y_1}{-1} = \frac{2(2h-k)}{5} \Rightarrow hk = 1$$

$12x^2 - 7xy - 12y^2 + 25 = 0.$

64. 5

Sol. Use Homogenization.

65. 3

Sol. Properties of G.I.F.

66. 0
Sol. Use L.M.V.T

SECTION – D

67. 02008.00

Sol. $t_n = \sum t_n - \sum t_{n-1}$

68. 01008.00
Sol. Use fundamental principle of counting.

69. 01210.00
Sol. (AM > GM)